



ELENA SERAFINO

PHYSICS MASTER'S CANDIDATE

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ABOUT ME

MSc Candidate in Physics specializing in Complex Systems and Big Data, expecting to graduate in Winter 2026/2027. Currently completing my thesis on financial network analysis, I tackle multidisciplinary challenges with a rigorous quantitative approach. Highly fluent in English, I combine machine learning, statistical modeling, and a strong passion for design. By focusing on both the analytical depth and the aesthetic form of data, I translate complex information into clear, impactful insights. I am seeking a data-driven role in an innovation-focused company where my skills and personal values can contribute to forward-thinking projects.

EDUCATION

MASTER'S DEGREE IN PHYSICS

Curriculum "Physics of Complex Systems and Big Data", current GPA: 29.82/30

University of Rome Tor Vergata | 2025 - present

BACHELOR'S DEGREE IN PHYSICS

Final Grade: 102/110

University of Rome Tor Vergata | 2021 - 2024

DIPLOMA DI MATURITÀ

Liceo Classico, final grade: 100/100

Locri (RC) | 2016 - 2019, Albano Laziale (RM) | 2019 - 2021

ACADEMIC PROJECTS

(Note: Full reports, methodologies, and presentations are available on my [portfolio](#))

GLOBAL VALUE CHAIN NETWORK ANALYSIS

Tech: Python, Network Science, FactSet Financial Data, Statistical Mechanics

- Engineered a macroscopic product-level economic network by merging, cleaning, and normalizing massive firm-level supply chain and revenue datasets (RBICS).
- Conducted topological and temporal persistence analysis (2018-2023) to identify the market's structural backbone, laying the groundwork for filtering out corporate diversification noise using maximum-entropy null models.

POWER GRID TIME SERIES FORECASTING

Tech: Python, Time Series analysis, Predictive models (SARIMA)

- Analyzed years of high-frequency power grid data to isolate macroeconomic trends.
- Developed a robust forecasting model to predict short-term energy demand and support load optimization.

SEISMIC RISK & STATISTICAL ANALYSIS

Tech: Python, Statistical Modeling, Extreme Value Theory

- Applied Extreme Value Theory (EVT) and Information Theory to model critical singularities and structural phase shifts in empirical datasets.

PARALLEL 2D ISING MODEL SIMULATION

Tech: C, MPI, High-Performance Computing

- Developed a parallelized Monte Carlo simulation to analyze complex system dynamics, evaluating algorithm efficiency and communication overhead on multicore architectures

SPEECH EMOTION RECOGNITION

Tech: Python, Machine Learning (SVM), Audio Processing

- Built a Machine Learning pipeline to classify human emotions from audio signals, processing complex temporal data to outperform standard baseline algorithms.

SKILLS

- Strong organizational skills, autonomy, and time management.
- High mental flexibility, curiosity, and cross-functional problem-solving.
- Tenacious, goal-oriented, and deeply committed to project success.
- Highly punctual with a strict focus on analytical and visual accuracy.
- Excellent written and verbal communication, with a natural aptitude for teamwork and leadership.
- Proficient in interacting with AI agents and prompting to optimize workflows and accelerate analysis.

TOOLS

- Python: Pandas, NetworkX, Data Libraries, Graphics Libraries
- C: MPI Parallel Computing
- Software & Design: LaTeX, Microsoft Office Suite, Canva, Github
- Languages: Italian (Native), English (Advanced)